

Quan Hua

Senior AI Engineer

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Professional Summary

Senior AI Engineer (10+ years) specializing in Computer Vision, deep learning, and production ML systems. Built production pipelines for 3D body measurement at BodiData (patent holder, 3 scanner generations) and autonomous robotics (CES 2015 TechCrunch Hardware Battlefield finalist). Founded AIPriceAction — Rust backend with 22 workers syncing 574 global tickers across stocks, crypto, and commodities. End-to-end ownership from model design to cloud/edge deployment.

Experience

Oct 2025 – Present: AIPriceAction - Creator & Lead Engineer

Website: aipriceaction.com GitHub: github.com/quanhua92/aipriceaction

- Engineered Rust backend (Axum, PostgreSQL, Redis) with 22 workers, syncing 574 global tickers across Vietnamese stocks, crypto, and global markets, enhancing market data coverage.
- Designed a multi-agent research pipeline in Python (Supervisor → Workers → Aggregator → Reviewer) with automated quality gates.
- Published 3 installable AI Agent Skills for Claude Code, Gemini CLI, Cursor, and OpenCode.
- Shipped a React 19 + TypeScript SPA (17 routes, 47K+ lines) with TradingView-style charts and backtesting playground.
- Built a Python SDK (aipriceaction) and CLI (aipa-cli) with AI analysis across 13 templates.

2017 – Oct 2025: BodiData, Inc - <https://www.bodidata.com/> Senior Machine Learning Engineer

3D body measurement and size-matching solutions to reduce clothing returns. Led the CV engine across 3 generations of the Kora® 3D Body Scanner.

- **Kora V1 (Surface + RealSense):** Designed the CV pipeline from scratch — integrated ORB-SLAM 3 for real-time camera positioning, implemented the C++ application in Qt. Collaborated with a PhD researcher on the radar algorithm.
- **Kora V2 (RealSense D415):** Ported to far-range camera. Decreased 3D measurement error by ~1 inch through enhanced model fitting against fused point cloud data, improving product accuracy.
- **Kora V3 (iPad + ESP32 → AWS Lambda):** Rewrote ESP32 firmware to multi-threaded ESP-IDF with BLE to iPad and Vayyar radar integration for depth fusion.
- **Kora V3 Cloud:** Refactored Qt C++ processing pipeline into a CLI binary and built a Python Lambda to orchestrate S3 downloads, processing, and uploads.
- Re-engineered Node.js SSE streaming server in Rust (Axum, Tokio), eliminating OOM crashes and boosting throughput 10x on smaller ECS instances.

2014 – 2017: Autonomous, Inc. - <https://autonomous.ai/> Artificial Intelligence Engineer

- Directed 6 AI engineers to build deep learning brain for Maya Personal Robot (CES 2015 TechCrunch Hardware Battlefield finalist), contributing to \$161K+ raised on Kickstarter.
- Integrated and deployed deep learning models for computer vision, facial recognition, emotion detection, and NLP as ROS components with real-time inference on edge hardware.
- Owned the ML pipeline from data collection to production inference via TensorFlow Serving and Flask across multiple product lines.

Patents

Co-inventor. “Systems and Methods for Generating a Refined 3D Model Using Radar and Optical Camera Data.” US20190295319A1, filed March 14, 2019.

Education

2010 – 2014: Bachelor of Science in Computer Science Honor Programs - University of Science Ho Chi Minh City. GPA: 8.7 / 10

Skills

- **Languages:** Rust, Python, C++, TypeScript
- **ML/DL:** PyTorch, Computer Vision, 3D Reconstruction
- **Frameworks:** LangChain, LangGraph, Qt, ESP-IDF, Axum, Tokio, React
- **Infrastructure:** PostgreSQL, Redis, Docker, AWS (Lambda, S3, ECS), CI/CD
- **AI Agents:** Multi-Agent Pipelines, Tool-Augmented LLMs, OpenAI API

Side Projects

UKYS: Great Fitting Shirts with Phone Measuring (Jun 2015 – Jan 2016)

Developed core CV body measurement algorithm for iPhone/iPad, enabling minute-long custom shirt sizing; won First Prize in The Ambassador’s Entrepreneurship Challenge — presented by US Ambassador to Vietnam.

Publications

Book: Machine Learning with TensorFlow 1.x (2017) Co-authored 5 of 13 chapters on Deep Learning with TensorFlow.

Book: OpenCV3 Blueprints (2015) Co-authored 2 chapters: “Recognizing Facial Expressions with Machine Learning” and “Panoramic Image Stitching Application using Android Studio and NDK.”

2014: Hong-Quan Hua, T. Hoang Ngan Le, and Bac Le, “An Effective Initialization for ASM-based Methods”, CISIM 2014 Conference.